

On the Security of the Automatic Dependent Surveillance-Broadcast Protocol

Martin Strohmeier, Vincent Lenders, and Ivan Martinovic

Abstract—Automatic dependent surveillance-broadcast (ADS-B) is the communications protocol currently being rolled out as part of next-generation air transportation systems. As the heart of modern air traffic control, it will play an essential role in the protection of two billion passengers per year, in addition to being crucial to many other interest groups in aviation. The inherent lack of security measures in the ADS-B protocol has long been a topic in both the aviation circles and in the academic community. Due to recently published proof-of-concept attacks, the topic is becoming ever more pressing, particularly with the deadline for mandatory implementation in most airspaces fast approaching. This survey first summarizes the attacks and problems that have been reported in relation to ADS-B security. Thereafter, it surveys both the theoretical and practical efforts that have been previously conducted concerning these issues, including possible countermeasures. In addition, the survey seeks to go beyond the current state of the art and gives a detailed assessment of security measures that have been developed more generally for related wireless networks such as sensor networks and vehicular *ad hoc* networks, including a taxonomy of all considered approaches.

Index Terms—ADS-B, aviation, air traffic control, air traffic management, NextGen, security, wireless, broadcast.

I. INTRODUCTION

THE WORLD of air traffic control (ATC) is moving from uncooperative and independent (primary surveillance radar, PSR) to cooperative and dependent air traffic surveillance (secondary surveillance radar, SSR). This paradigm shift holds the promise of reducing the total cost of deployment and improving the detection accuracy of aircraft. However, it is well known in the aviation community that the ATC system, which is currently being rolled out, called *automatic dependent surveillance-broadcast* (ADS-B), has not been developed with security in mind and is susceptible to a number of different radio frequency (RF) attacks. The problem has recently been widely reported in the press [1]–[5] and at hacker conventions [6]–[8]. Academic researchers, too, proved the ease of compromising the security of ADS-B with current off-the-shelf hard- and software [9]. This broad news exposure led the International Civil Aviation Organization (ICAO) to put the security of civil aviation on their agenda of the 12th air

navigation conference, identifying “cyber security as a high-level impediment to implementation that should be considered as part of the roadmap development process” [10] and creating a task force to help with the future coordination of the efforts of involved stakeholders.

This shows that there is a widespread concern about the topic, created by the fact that ADS-B will be mandatory for all new aircraft in the European airspace by 2015¹ and has already been embraced by many airlines worldwide. Reports from manufacturers and regulation bodies show that around 70%–80% of commercial aircraft worldwide have been equipped with ADS-B transponders as of 2013 [11], [12]. Countries such as Australia have already deployed full continental coverage, with ADS-B sensors being the single means of ATC in low population parts of the country [13].

This paper gives an overview of the research that has been conducted regarding the security of ADS-B and describes the potential vulnerabilities identified by the community. Since much relevant security research has been conducted in related fields such as wireless sensor networks or general *ad hoc* networks, we analyze proposed countermeasures from other areas that could be adapted for use in ADS-B or whether they are not applicable for reasons inherent to the system. Furthermore, the present survey provides a threat catalogue, analysis and vulnerability categorization of the mainly used data link *Mode S*. We focus primarily on the security of ADS-B and not on other air traffic control (sub-) systems, such as GPS. Among other questions, we seek to answer why existing ideas for securing (wireless) networks such as traditional cryptography cannot simply be transferred and used in the protection of systems such as ADS-B.

While there were a number of reasons behind the switch to a modern air traffic management system, cost has consistently been mentioned as one of the most important ones throughout the process; existing radar infrastructures are simply much more expensive to deploy and maintain [14]. ADS-B, on the other hand, provides significant operational enhancements for both airlines and air traffic managers. The increased accuracy and precision improves safety and decreases the likelihood for incidents by a large margin, unless the system’s weak security is exploited by malicious in- and outsiders.

The remainder of the survey is organized as follows. Section II gives a detailed overview over the security problems related to ADS-B and the requirements of the system environment. Section III outlines solutions proposed in previous works

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¹Older aircraft need to be retrofitted by 2017. The FAA mandates ADS-B in the US airspace by 2020.

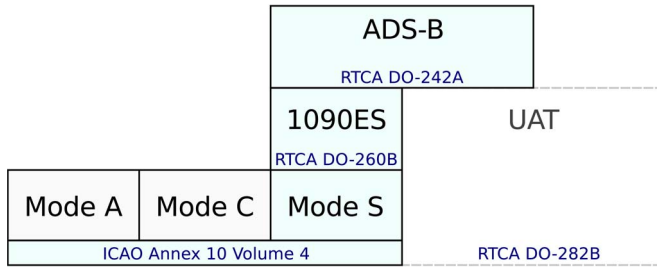


Fig. 1. ADS-B hierarchy [9]. The 1090 MHz Extended Squitter is based on the traditional Mode S system and provides the data link for ADS-B in commercial aviation. UAT is a new development but currently only mandated for general aviation.

and looks at the sister protocol ADS-C and military versions of ADS-B. Section IV surveys secure broadcast authentication methods to address the problem while Section V reviews means to establish secure location verification with ADS-B. Section VI summarizes and Section VII concludes the survey.

II. PROBLEM STATEMENT

This section defines the problems related to security in ADS-B more thoroughly. First, we give a short overview over the currently used ADS-B protocol and its existing vulnerabilities. Building on this, a model of the ADS-B environment is outlined and the required attributes of possible solutions are identified.

A. ADS-B Protocol Overview

The American Federal Aviation Administration (FAA) as well as its European pendant EUROCONTROL named ADS-B as the satellite-based successor of radar. At its introduction, ADS-B was a completely new paradigm for air-traffic control. Every participant retrieves their own position and velocity by using an onboard GPS receiver. The position is then periodically broadcasted in a message (typically twice per second) by the transmitting subsystem called ADS-B Out. The messages are then received and processed by ATC stations on the ground as well as by nearby aircraft, if equipped with the receiving subsystem ADS-B In. Messages can integrate further fields such as ID, intent, urgency code, and uncertainty level.

Two competing ADS-B data link standards have been proposed: Universal Access Transceiver (UAT) and 1090 MHz Extended Squitter (1090ES). UAT has been created specifically for the use with aviation services such as ADS-B, utilizing the 978 MHz frequency with a bandwidth of 1 Mbps. Since UAT requires fitting new hardware, as opposed to 1090ES, it is currently only used for general aviation in EUROCONTROL and FAA-mandated airspaces. Commercial aircraft, on the other hand, employ SSR Mode S with Extended Squitter, a combination of ADS-B and traditional Mode S known as 1090ES (see Fig. 1). In other words, the ADS-B function can be integrated into traditional Mode S transponders. From here on, we focus on the commercially used 1090ES data link. The complete overview over the ADS-B protocol can be found in the specification documents [15]–[17] while various other works give succinct, higher level descriptions of the protocol (e.g., [6], [9], and [18]; see Fig. 2).

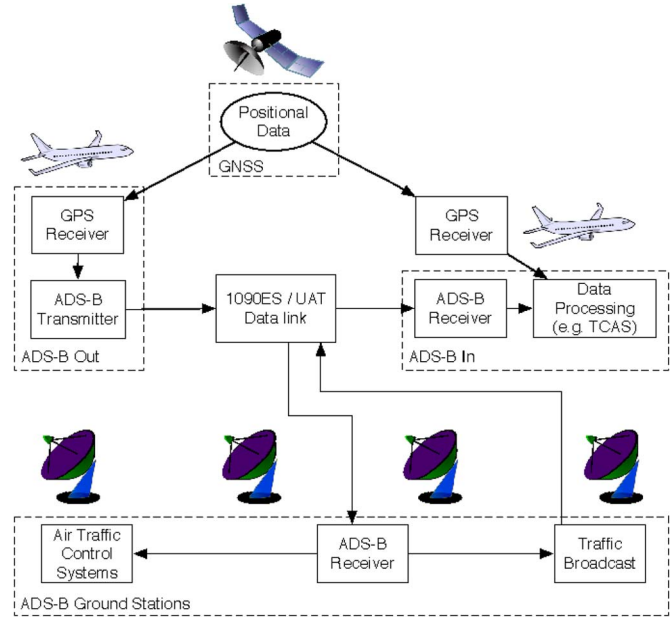


Fig. 2. Overview of the ADS-B system architecture. Aircraft receive positional data that is transmitted via the ADS-B Out subsystem over the 1090ES or the UAT data link. It is then received and processed by ground stations and by other aircraft via the ADS-B In subsystem.

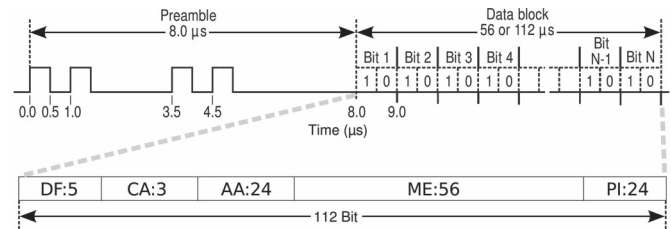


Fig. 3. 1090 ES data link [9].

The 1090ES Data Link: As the name suggests, the 1090ES data link predominantly uses the 1090 MHz frequency for communication sent out by aircraft, to both other aircraft and ground stations (Mode S also uses ground to air communication at 1030 MHz for interrogations and information services). Fig. 3 provides a graphical view of a 1090ES transmission, which starts off with a preamble of two synchronization pulses. The data block is then transmitted by utilizing pulse position modulation (PPM). With every time slot being 1 μ s long, a bit is indicated by either sending a 0.5 μ s pulse in the first half of the slot (1-bit) or in the second half (0-bit). It is important to note that PPM is very sensitive to reflected signals and multipath dispersion, a fact that can play a major role in security and protocol considerations.²

There are two different possible message lengths specified in Mode S, 56 bit and 112 bit [15], whereas ADS-B solely uses the longer format. The downlink format field DF (alternatively UF for uplink messages) assigns the type of the message. 1090ES uses a multipurpose format, as shown in Fig. 3. When set to 17, it indicates that the message is an extended squitter, enabling the transmission of 56 arbitrary bits in the ME field. The CA field indicates information about the capabilities of the

²See [19] for more information on PPM and multipath.

TABLE I
COMPARISON OF CIVIL AVIATION TRANSPONDER MODES [22]

	Message Length	Frequencies	Operational Mode	Use Cases
Mode A	12 bit	1030 / 1090 MHz	Independent / Non-selective interrogation	Identification
Mode C	12 bit	1030 / 1090 MHz	Independent / Non-selective interrogation	Pressure Altitude Extraction
Mode S	56 / 112 bit	1030 / 1090 MHz	Independent / Selective interrogation	Multiple
ADS-B / 1090 ES	112 bit	1090 MHz	Dependent / Automatic	Multiple

employed transponder, while the 24 bit AA field carries the unique ICAO aircraft address which enables aircraft identification. Finally, the PI-field provides a 24 bit CRC to detect and correct possible transmission errors. It is possible for recipients to correct up to 5 bit errors in 1090ES messages using a fixed generator polynomial of degree 24.

This quick overview shows that only the 56 bit ME field can be used to transmit arbitrary data, i.e., utilized for a secure ADS-B solution. However, not only is it very limited in size but it is also typically occupied by positional and other data. Thus, the format as currently in practical use is intuitively very limiting to most types of security solutions as we will explain in more detail in this survey.

B. Relation to Legacy Systems

Traditionally, aircraft localization has been relying on radar systems which had been developed for military applications, namely identification, friend or foe (IFF) systems. There are two different concepts in conventional radar: primary surveillance radar and secondary surveillance radar [20]. PSRs are *independent*; they work without cooperation from the aircraft by transmitting high-frequency signals, which the target object reflects. The echo identifies range, angular direction, velocity, size and shape of the object. SSR, on the other hand, uses interrogations from ground stations which are responded to by transponders in aircraft. The reply includes information such as the precise aircraft altitude, identification codes or information about technical issues. In contrast to PSR, this approach is also much more accurate in terms of localization and identification. As all surveillance data such as position and status are derived directly by the aircraft, SSR is *dependent*. Furthermore, cooperation by the aircraft is a requirement.

Before ADS-B, all SSR systems in ATC have been interrogation-based. So called *modes* are being used to query the identification and altitude of an aircraft. There are three modes (A, C, and S) currently in use in civil aviation, Table I compares their characteristics with ADS-B. The latter embodies a paradigm shift in ATC as air traffic surveillance is now cooperative and dependent, i.e., every aircraft collects their own data such as position and velocity by using onboard measurement devices. ADS-B based surveillance infrastructure is also much more cost-effective compared to conventional PSR, which is a much more complex technology that also suffers from wear and tear due to rotating parts. ICAO specifies the technological

cost of using primary radar to monitor an en-route airspace (200 NM radius) at \$10–14 million, while Mode S surveillance is priced at \$6 million and ADS-B is significantly cheaper at \$380 000 [21].

C. ADS-B Vulnerabilities

In this section, we discuss the ADS-B vulnerabilities inherently stemming from the broadcast nature of RF communication when used without additional security measures. Contrary to wired networks, there are no practical obstacles for an attacker trying to access a wireless network such as buildings or security guards, making access control mechanisms very challenging. In [18], McCallie *et al.* defined a taxonomy for various possible attacks, even though their difficulty estimations have become somewhat dated with the widespread availability of cheap software-defined radios as recently illustrated in [9]. Despite this, the described attacks are ordered in increasing order of difficulty here, providing a comprehensive attacker model in the context of ADS-B vulnerabilities (see Table II).

Eavesdropping: The most straightforward form among the many security vulnerabilities present in ADS-B is the act of listening in to the unsecured broadcast transmissions. This passive attack is called *Aircraft Reconnaissance* in [18]. As ADS-B is using unsecured messages over an inherently broadcast medium, the possibility to eavesdrop is not surprising and has been mentioned since the early stages of development. Many non-adversarial services use this obvious privacy concern, e.g., to visualize air-traffic on the Internet,³ yet eavesdropping also forms the basis for a number of more sophisticated active attacks. Furthermore, eavesdropping is not only difficult to prevent without applying full encryption but it is also practically impossible to detect. A small number of countries (such as the United Kingdom) have long-standing, very general laws against listening in on unencrypted broadcast traffic which is not intended for the recipient⁴ even though the technical realities render such legal approaches all but obsolete.

Jamming: Almost equally simple is the jamming attack, where a single node (both ground stations or aircraft) or an area with multiple participants is effectively disabled from sending/receiving messages by an adversary sending with sufficiently high power on the 1090 MHz frequency of Mode S. It has generally also been proven feasible to do reactive jamming in real time, targeting only packets which are already in the air as assessed in [23]. While jamming is a problem common to all wireless communication, the impact is severe in aviation due to the system's inherent wide open spaces which are impossible to control as well as the importance and criticality of the transmitted data. Besides SSR communications systems such as ADS-B, primary radar may also be a target of jamming attacks

³Prominent examples are flightradar24.com and radarvirtuel.com among many others.

⁴Section 48 of the Wireless Telegraphy Act of 2006 states that (1) "A person commits an offence if, otherwise than under the authority of a designated person—(a) he uses wireless telegraphy apparatus with intent to obtain information as to the contents, sender or addressee of a message (whether sent by means of wireless telegraphy or not) of which neither he nor a person on whose behalf he is acting is an intended recipient, or (b) he discloses information as to the contents, sender or addressee of such a message."

TABLE II
OVERVIEW OF VULNERABILITIES IN THE ADS-B PROTOCOL. THE TABLE SUMMARIZES THE ATTACKS INCLUDING SEVERITY AND COMPLEXITY

Attacks	Layer	Method	Severity	Complexity	References
Aircraft Reconnaissance	PHY+APP	Eavesdropping	Low	Lowest	[6], [9], [18]
Ground Station Flood Denial	PHY	Signal Jamming	Medium	Lower	[6], [9], [18], [23], [24]
Aircraft Flood Denial	PHY	Signal Jamming	Medium	Low-Medium	[6], [9], [18], [23], [24]
Ground Station Target Ghost Injection / Flooding	APP	Message Injection	High	Low	[6], [9], [18]
Aircraft Target Ghost Injection / Flooding	APP	Message Injection	Medium	Low-Medium	[6], [9], [18]
Virtual Aircraft Hijacking	PHY+APP	Message Modification	High	Medium	[9], [25], [26]
Virtual Trajectory Modification	PHY+APP	Message Modification	High	Medium	[9], [25], [26]
Aircraft Disappearance	PHY	Message Deletion	High	Low	[9], [18]
Aircraft Spoofing	PHY+APP	Message Modification	High	Low	[6], [9], [24]

which is similar in many ways, most importantly the fact that one always jams the receiver (in our case ATC systems), not the transmitter.⁵ A thorough introduction to radar and communication jamming can be found in [27].

Due to rotating antennas and a higher transmission power, typical PSRs are more difficult to jam than ADS-B receivers, especially for non-military grade attackers. However, as there are usually many distributed ADS-B receivers for ATC purposes, it still takes considerable effort to completely blackout a given area. Notwithstanding this, a targeted attack would create major denial-of-service problems at any airport. Jamming moving aircraft is also possible, however considered more difficult. In summary, jamming is integral for the following attacks [18]:

- Ground Station Flood Denial.
- Aircraft Flood Denial.

Message injection: On the next higher level of difficulty, it is also possible to inject non-legitimate messages into the air-traffic communication system. Since no authentication measures are implemented at the data link layer, there is no hurdle at all for an attacker to build a transmitter that is able to produce correctly modulated and formatted ADS-B messages. See [9] for more details on how to conduct an attack with limited knowledge and very cheap and simple technological means which have been easily and widely available for some time. As another direct consequence of missing authentication schemes, a node can deny having broadcasted any (false) data and/or claim having received conflicting data, making any kind of liability impossible. Concrete attack instances that use message injection include [18]:

- Ground Station Target Ghost Injection/Flooding.
- Aircraft Target Ghost Injection/Flooding.

Message deletion: Legitimate messages can be physically “deleted” from the wireless medium by utilizing destructive or constructive interference. Destructive interference means transmitting the inverse of the signal broadcast by a legitimate sender. Due to superposition, the resulting signal should be

erased or at least highly attenuated but in practice this approach has very precise and complex timing requirements, making it extremely challenging.

Constructive interference on the other hand does not require synchronization but simply causes a large enough number of bit errors. Since Mode S extended squitters’ CRC can correct a maximum of 5 bit errors per message, if a message exceeds this threshold, the receiver will drop it as corrupted. While effectively destroyed, the receiver might at least be able to verify that a message has been sent, depending on the implementation and the circumstances. In any case, it is more subtle than complete jamming of the 1090 MHz frequency. Besides providing an easy means of message modification in conjunction with message injection, message deletion is key to the following attack:

- Aircraft Disappearance.

Message modification: Modifying messages on the physical layer during transmission is typically done via two different approaches, overshadowing and bit-flipping. Overshadowing means that the attacker sends a high-powered signal to replace part or all of the target message. Bit-flipping on the other hand has the attacker superimposing the signal converting any number of bits from 1 to 0 (or the other way around). In both cases arbitrary data can be injected without the knowledge of any of the participants. This effect can also be achieved by combining message deletion and injection, but physical layer message modification can in some cases be regarded as even more sinister than the injection of a completely new message, since the manipulated message was originally legitimate. The feasibility of such message manipulation has recently been shown in [25] and [26]. Concrete attack examples are [9]:

- Virtual Aircraft Hijacking.
- Virtual Trajectory Modification.

D. Identification of System Requirements

There are a number of demands on a security approach for ADS-B, stemming from the way it is needed to work in practical real-world aviation settings and the characteristics of the broadcast approach. We first codify the model and then

⁵Of course, PSR does have both a transmitter and a receiver, while ADS-B ground stations are dependent on the tracked targets’ transmissions.

follow up with an analysis of what we need to have as security primitives in air traffic control systems.

1) *Network Properties:*

- The assumed network model consists purely of unidirectional broadcasts. Although there is a growing body of research on Aeronautical Ad hoc Networks (AANETs) that provide multi-hop communication [28], the present real-world implementation is based on **single-hop unidirectional broadcast links**. Aircraft broadcast their position, velocity and direction in plain-text periodically every few hundred milliseconds, a concept called beaoning. Thus, in the following we concentrate mainly on the so-called beacon-based security.
- **We do not consider any type of energy constraints** in association with ADS-B devices.
- Furthermore, there are **no significant computational constraints** with neither ground stations nor ADS-B units employed in aircraft.
- **Reliability has not been a major concern** yet, as some lost packets do not normally cause a problem. Indeed, the protocol does have no means to prevent collisions, the sender will not retransmit any packets and no guarantees are given. Loss is dealt with on higher layers. This is also reflected in [9], which shows that the packet error rate tends to hover around a mean of 33%, independent of the channel. This means that there is some substantial packet loss on the physical layer that is likely to increase when the channel utilization rises over the next decades due to the mandatory ADS-B roll-out. This will be reinforced by the ever-increasing flight traffic,⁶ especially in high-density airspace.
- *The network is ad hoc and highly mobile.* Many nodes are moving at a velocity of up to 1000 km/h or more. It is therefore extremely dynamic and communication between two nodes may last only a few seconds. Aircraft trajectories are principally not physically restricted, although there are often common routes and also some air spaces that are restricted due to policies.
- **The network is long range**, typically ADS-B is considered to be feasible at distances of 100 NM and more.⁷
- In contrast to other, independently deployed wireless (sensor) networks the **undetected physical capture of legitimate nodes is not the most important concern**. Legally having access to a legitimate ADS-B node, however, is not considered very difficult, at least when taking general aviation into account.

2) *Security Attributes:* Perrig and Tygar [30] identify two large themes around which secure broadcast revolves: First, make sure that receivers know that any received information comes from the appropriate sender, and second that senders can freely limit the recipients of broadcasted information. Confidentiality, e.g., the protection of ADS-B position messages to prevent attacks in air or loss of trade secrets, has not been

considered in the development of the system as laid out in a formal security requirements engineering for ADS-B in [31].

In light of the recently exploited inherent vulnerabilities of the ADS-B system with cheap off-the-shelf hardware [6], [9], this is quickly becoming the most important topic in ADS-B research. Based on the model defined above, any security scheme for ADS-B would have to satisfy the following properties:

- *Data integrity:* Ensures that the data are the same as has been provided by the sender and has not been modified by any third party.
- *Source integrity:* Ensures that a message originates from the participant that claims to have sent it.
- *Data origin authentication:* Ensures that a message originates from the location claimed in a message.
- *Low impact on current operations:* A scheme should be compatible with the current ADS-B installations and not overly affect both hard- and software standards.
- Sufficiently *quick and correct detection* of incidents.
- Needs to be *secure against DoS-attacks* against computing power.
- Any approach *needs to be easily scalable*. This is in respect to both a locally rising aircraft density and globally increasing aircraft traffic. The strain on the heavily used 1030 MHz channel should not measurably increase (e.g., due to an increased number and/or larger packets).
- *Robustness to packet loss.* A jammed wireless channel should decrease neither security nor reliability of the scheme.
- Achieving *non-repudiation* is seen as nice to have but not very high on the priority list for immediate air traffic security and is more of a legal topic.

III. OUTLINING SOLUTIONS

Substantial work has already been done on ADS-B security over the last decade and several approaches have been proposed in the literature to enhance ADS-B security in particular. Furthermore, a large amount of research has been done in related fields such as vehicular *ad hoc* networks (VANETs) and wireless sensor networks where broadcast authentication and security also play an important role. While some ideas may not be directly useful to the aforementioned requirements of ADS-B, it might be possible to adapt them. We list the major obstacles for their application, along with the advantages and disadvantages of each approach.

The remainder of this section first presents an overview of works in the community concerned with the security of ADS-B as a whole. We then consider ADS-B's sister protocols ADS-C and military versions of ADS-B in our context before analyzing all collected ideas in detail in the following sections.

As shown in the taxonomy in Fig. 4, we identified two distinct approaches to securing ADS-B: **Secure Broadcast Authentication** and **Secure Location Verification**. Consequently, Section IV examines the various schemes that apply asymmetric properties (cryptographic and non-cryptographic) to directly authenticate broadcast communication while Section V reviews several different methods that seek to verify the authenticity of location claims made by aircraft and other ADS-B participants.

⁶Growth rate forecasts from market analysts suggest a 5.1% increase per year between 2010 and 2030. Cargo traffic is expected to grow even more at 5.6% per year in the same time frame [29].

⁷Aviation typically uses the nautical mile, also abbreviated as nmi, as a distance measure which is 1852 m.

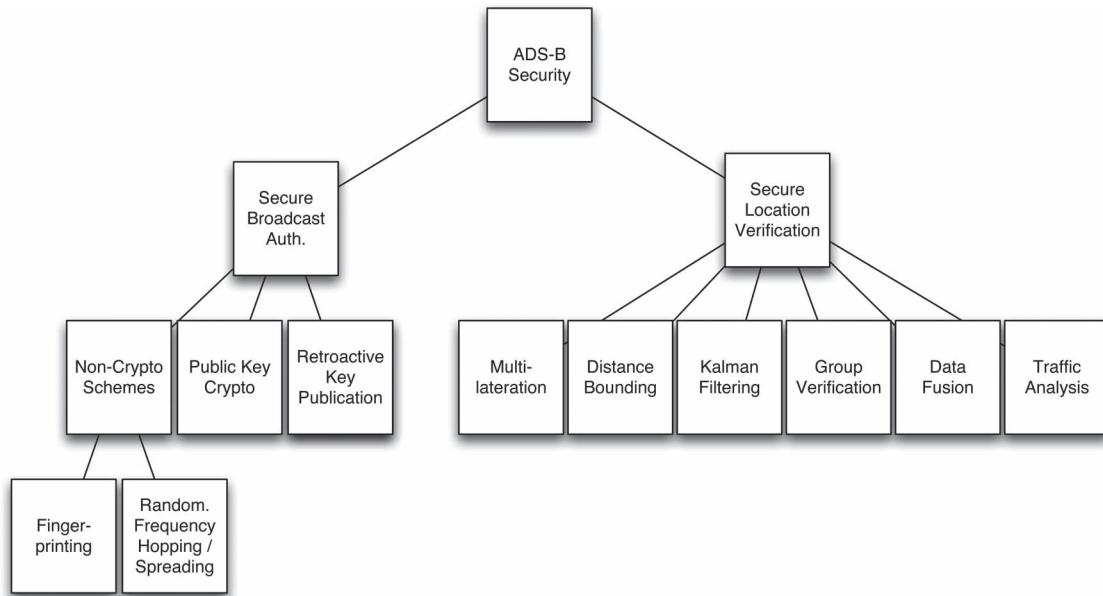


Fig. 4. Taxonomy of ADS-B security.

A. Previous Works on ADS-B Security

Securing ADS-B communication was not a very high priority when it was specified to be the new standard in civilian secondary surveillance. Neither the official standards of the Radio Technical Commission for Aeronautics (RTCA) [15]–[17] nor other requirements documents [32], [33] mention security in this context. However, security problems in ADS-B have been well-known for a long time, mostly because they are relatively obvious to the interested researcher.⁸ For instance, weak ADS-B security has recently got very broad reporting in the mainstream press⁹ due to two talks at the DEFCON and Black Hat security conferences.¹⁰

Sampigethaya and Poovendran [34], [35] first analyze the security and privacy of ADS-B and other related and unrelated communication systems which are part of the so-called “e-enabled aircraft”. McCallie *et al.* [18] provide a current security analysis, focused on the nature of possible attacks and their difficulty. They give a systematic high-level overview and propose general recommendations for addressing ADS-B’s problems. Costin and Francillon [6] as well as Schaefer *et al.* [9] analyze ADS-B security, too, focusing on the ease of exploiting ADS-B with current hard- and software and offering some possible countermeasures.

Kovell *et al.* [36] mention data fusion with other systems, multilateration and various cryptographic schemes as state of the art research on ADS-B security. They further conduct a more thorough analysis on Kalman filtering and group validation concepts and proposed mitigation methods. Nuseibeh *et al.* conduct an example of a formal security requirements

⁸Especially on the Internet broad warnings have been floating around as early as 1999, e.g., <http://www.airsport-corp.com/adsb2.htm>, http://www.dicksmithflyer.com.au/cat_index_36.php

⁹E.g., <http://www.forbes.com/sites/andygreenberg/2012/07/25/next-gen-air-traffic-control-vulnerable-to-hackers-spoofing-planes-out-of-thin-air/>

¹⁰<http://www.blackhat.com/usa/bh-us-12-briefings.html#Costin>

analysis of ADS-B [37], proposing multilateration to deal with possible attack scenarios. Burbank *et al.* [38] present general concepts for communications networking to meet the requirements of future airspace systems, i.e., a vision of a mobile *ad hoc* and wireless networking concept for use in both the terminal area and in the en-route airspace. Li and Kamal [39] analyzed the security of the whole FAA’s Next Generation Air Transportation System (NextGen) of which ADS-B is a core component.¹¹ They develop a high-level defense-in-depth framework for analyzing NextGen and mention general secure communication approaches such as encryption, authentication and spread spectrum as part of a possible ADS-B security layer that would need to be examined more deeply.

The number of works relating to the security of ADS-B has been increasing steadily as the final mandate for its use in US and other Western airspaces is drawing closer and the problem is becoming more urgent. While the existing works pertain the need for security and offer insights into different aspects, especially multilateration and data fusion, the present work seeks to widen the understanding of the communications community and address the problem on a much broader and more comprehensive scale. We look to compile all previously considered aspects, seek to include some applicable, overlooked ideas from other areas of network security, and weigh the advantages and disadvantages of the examined approaches against each other.

B. ADS-C

A theoretical, already available, way to deal with suspicious ADS-B participants would be to ask them to switch to the connection-oriented ADS-C (ADS-Contract, also known as ADS-Addressed). However, ADS-C also has a number of very

¹¹Although certainly not the only safety-critical wireless module. See, e.g., [40] for concerns about the impact of GPS integrity on aviation safety.

severe inherent shortcomings, some of which are described by ICAO: [41]

- Additional avionics systems (for data communications) such as the Future Air Navigation System (FANS) 1/A or the Aeronautical Telecommunications Network (ATN) are needed.
- Its performance may be limited by the communications medium.
- Currently, ADS-C data are carried by a data link service provider, so a cost may be incurred for each transmission.
- Unlike the ADS-B IN system, ADS-C messages are not directly available to other aircraft.

Overall, the current implementation of ADS-C over the outdated ACARS network¹² considerably limits its usefulness. It is also violating the basic requirements of ADS-B as it is a system that sends on demand instead of periodically broadcasting without being requested. Connection-oriented protocols lack most of the advantages that the paradigm change with ADS-B provides to modern ATC (specifically cost, scalability and ease of use), which is why ADS-C has not been considered any further in the development of NextGen.

C. Communication in Military Avionics

There is undoubtedly a much stronger need and motivation to implement stringent security in a military communications context. Though it is not the primary focus of this survey, anything in practical use by military forces could naturally be of interest for civil security solutions as well. There are various standards currently in use by the US and NATO military, among them the cryptographically secured Mode 4 and Mode 5 as defined in the NATO Standardization Agreement (STANAG) 4193. Mode 4, which employs a 3-pulse reply to a challenge, has been in use for decades and according to the forecasts of the NATO Minimum Military Requirements is to be superseded by Mode 5 in 2015 (initial operational capability) and 2020 (full operational capability), respectively [42].

While the legacy Mode 4 indeed only allows airplanes to respond to challenges, Mode 5 adopts the ADS-B broadcast capability, so participants can announce their presence without a prior query, very useful in identification, friend or foe [43]. On the security side, Mode 5 uses proprietary hardware and encryption algorithms with a black key concept;¹³ it furthermore offers time-of-day authentication, automatic switchovers and a longer period [44]. The signal modulation is done via spread spectrum and operation requires a platform identification number (PIN). Mode 5 hardware is equipped with a unique identifier that informs about national origin and the platform number. Mode 5 has two different levels: Level 1 is the interrogation response mode, providing time, position and identification based on both GPS and traditional means. Level 2 is the broadcast mode and entirely based on GPS. There are currently no further available

¹²The Aircraft Communications Addressing and Reporting System (ACARS) will be superseded by the Aeronautical Telecommunications Network (ATN) and IP communication over the next decade.

¹³Black keys are safe to transmit since they are encrypted with an encryption key. Red keys on the other hand are unencrypted and classified as highly sensitive.

Format #	DF				
19	1 0011	AF : 3	Military Application : 104		Military Extended Squitter

Fig. 5. DF19 data format [45].

details on the security mechanisms including the applied cryptography of Mode 4 and 5 since this information is classified.

The ADS-B specification itself also mentions the message types/downlink formats Military Extended Squitter (DF19, see Fig. 5) as well as Military Use Only (DF22) without detailing them further, although it is known for example that DF19 makes ample use of bursts instead of regular beacon messages only [45]. Despite incomplete or unavailable information on the performance of Mode 5 compared to ADS-B, it is safe to say that both cost, scalability and ease of use of the known aspects of the system are prohibitive to widespread use in commercial ATC. Spread spectrum techniques and cryptography could, however, be a part of a future security approach to ADS-B and will be discussed in this survey.

IV. SECURE BROADCAST AUTHENTICATION

Secure Broadcast Authentication is one possible means to prevent and/or detect attacks in a unidirectional broadcast network such as ADS-B. This section will describe the various methods that have been proposed in the literature, typically for wireless sensor networks or VANETs and analyze their applicability to ADS-B.

Authentication of messages on a broadcast medium is hard, compared to point-to-point communication. A symmetric property is only useful in point-to-point authentication where both parties trust each other. Thus, an asymmetric mechanism is inherently required so that receivers can *verify* messages but are not able to *generate* authentic messages themselves [46]. For a good overview over secure broadcast communication in general, the reader is referred to [30].

The goal is to keep the open nature of ADS-B intact while offering a potential authentication mechanism. This could be done either globally or only selectively in cases where suspicious behavior has been detected. Such reactive authentication could lessen the strain on the network by only requiring additional security (and thus computational and communicational overhead) at times when incidents seem more likely.

Furthermore, there is a distinction between broadcast schemes that are user-based vs. those that are node-based, or possibly both. Node-based (also known as host-based) schemes ensure the authenticity of a given node, i.e., the hardware. User-based schemes on the other hand look to authenticate a human user, regardless of the underlying hardware [47]. This survey focuses mainly on node-based schemes.

A. Non-Cryptographic Schemes on the Physical Layer

Non-cryptographic schemes such as fingerprinting comprise various methods for wireless user authentication and device identification techniques, either based on hardware or software imperfections or characteristics of the wireless channel which are hard to replicate. The goal is to identify suspicious activity in a network. Finding a signature for legitimate beacons in

a network, possibly being able to tell apart ground stations from aircraft, identifying the type of aircraft or even individual machines provides data useful for the development of an intrusion detection system [48]. If there are tangible differences between legitimate and non-legitimate packets on the physical layer, then machine learning techniques could be employed to develop a model for predictions of normal behavior and also statistical thresholds beyond which an activity is considered suspicious. Even if it is only feasible to identify classes of devices instead of singular participants, this could prove to be valuable information in detecting intruders. Yet, fingerprinting does not provide surefire security in any way, and various attacks and concerns have been brought forward [49].

Currently, there have been no attempts at applying any kind of non-cryptographic schemes to boost the security of ADS-B. A common counter-argument has been the fact that contrary to, e.g., the 802.11 markets the commercial airplane market is divided into two big players (Boeing and Airbus) which in the long run makes significant differences at least between ADS-B vendors unlikely. Still, fingerprinting has also been successfully employed to tell apart the exact same models from the same vendor. For a good overview of the state of the art in physical-layer identification of wireless devices, see [48].

Zeng *et al.* [50] broadly identified three different techniques that can be employed to enhance or even replace traditional cryptographic measures: Software, hardware and channel-based fingerprinting.

a) Software-based fingerprinting: This type of fingerprinting techniques tries to exploit distinctly different patterns or behavior of software operating on wireless equipment. Depending on the specification of a protocol, there is a lot of leeway for manufacturers and developers when implementing software on a given device. If there is enough entropy in information about the combination of chip sets, firmware, drivers to tell apart different wireless users, this approach can be used to verify their continuity up to a certain degree. As a downside, it seems likely that large fleets of airline operators are fitted with very similar or same hardware, making them harder or even impossible to differentiate and on the other hand easier to study and copy for a potential attacker.

b) Hardware-based fingerprinting: A number of techniques have been proposed to identify devices based on unique hardware differences. Some of these differences can be used for *radiometric fingerprinting*, exploiting differences in the turn-on/off transient (see, e.g., [51]) or the modulation of a radio signal to build unique signatures. While this works well for non-mobile cases and attackers with standard, off-the-shelf hardware, it can break down against more powerful adversaries employing software defined radios and be subjected to signal/feature replay attacks [48]. Furthermore, the existing research captured signals very closely to the fingerprinting antenna (15 m or less) and in non-mobile settings, making it very improbable to work in the highly-dynamic, large-distance ADS-B setting.

Another unique hardware feature amongst wireless devices is *clock skew*. As no two clocks run precisely the same, this can be used to create signatures and enable identification. Unfortunately, to exploit this, we would require timestamps included in ADS-B messages. Also, it is possible for an attacker

to eavesdrop on the communication and mimic the appropriate clock skew [52].

Recently reviewed options for future systems include the use of so-called physically unclonable functions (PUFs), which essentially exploit specifically implemented circuits to create unique and secure signatures, thus abandoning the scope of non-cryptographic solutions.¹⁴ Furthermore, besides requiring new hardware, this approach also necessitates an overhauled messaging protocol, including a challenge and response model [54], making it a difficult fit for the requirements of the ADS-B protocol.

c) Channel/location-based fingerprinting: Exploiting natural characteristics of the physical layer has been a hot research topic in relation with security in wireless networks. Various approaches have shown that this can be a viable alternative to more traditional authentication and verification measures, typically based on received signal strength (RSS, e.g., [55]), channel impulse response (CIR, e.g., [56]) or the carrier phase (e.g., [57]). They are comparably easy to implement in wireless systems and can offer reasonable security without requiring much overhead.

Any such concept requires bidirectional communication, however. One practical example is the retroactive authentication of data packets via an RSS list as proposed by Zeng *et al.* [50]. As this temporal RSS variation authentication (TRVA) requires an ACK packet in a given coherence time, it is not compatible with current ADS-B protocols. Furthermore, it is doubtful if this could work with reasonable efficiency in a highly dynamic environment such as airborne MANETs. The coherence time T_C in which the channel stays stable in a wireless network where only one sender moves at 800 km/h is roughly 0.6188 ms. At a 1090ES bandwidth of 1 symbol/ μ s it is obvious that such protocols are impossible to deploy. This physical property also effectively denies the application of many other more sophisticated physical-layer schemes such as SecureAngle [58], which aims at securing wireless networks by using multiple antennas capturing the angle of arrival information of nodes to build signatures and detect anomalies. Similarly, practical indoor location-based geo-tags, built with surrounding radio frequency signals such as done in [59] are not applicable.¹⁵

Laurendeau and Barbeau [60], [61] exploit RSS in a way similar to time difference of arrival concepts (see Section V-A) to localize malicious insiders in a vehicular *ad hoc* network with the help of various receivers. Despite the fact that an attacker will not be inclined to cooperate and can even actively fake the signal strength he utilizes, their proposition enables the receivers of a message to at least identify a given area where it must have originated from.

d) Randomized/uncoordinated frequency hopping/spreading: A physical-layer scheme different from fingerprinting, Frequency Hopping Spread Spectrum (FHSS) and Direct Sequence Spread Spectrum (DSSS) are both used in wireless systems to

¹⁴For a good overview on PUFs, see [53].

¹⁵Although one advantage is that many legacy Mode S systems use several directional rotating antennas to pick up ADS-B signals. The extracted information about the angle-of-arrival could conceivably be used to raise red flags, as described in Section V-F.



Fig. 6. Uncoordinated frequency hopping after [62]. Both A and B regularly change their communication frequencies without having pre-established a common pattern. By statistical chance they will communicate on the same channel every so often.

improve protection against malicious narrow band and pulse jamming as well as eavesdropping. In their usual form they both require a pre-shared spreading code or hopping pattern between sender and receiver which makes it hard to follow or hinder the communication for anyone without access to the code/pattern. This is also exploited in military communications (see Section III-C) but is not a viable option for world-wide civil and commercial ATC where such secret codes would presumably not stay secret for long.

The need for a pre-established code can be relinquished by employing random, uncoordinated versions of FHSS and DSSS. Strasser *et al.* [62] propose such a physical layer approach to counteract jamming in wireless broadcast scenarios. Uncoordinated Frequency Hopping (UFH) provides a viable way to broadcast initial messages without an attacker being able to jam the transmission in an efficient way. The key insight to these approaches is that, contrary to normal frequency hopping mechanisms, sender and receiver(s) rely on the statistical chance to be on the same channel at the same time (see Fig. 6). The obvious downside of UFH is its low bandwidth due to the fact that many times receivers will not listen on the correct channel. More concretely, the probability with UFH that a packet will be received at a node without an attacker being involved is $p_m \geq 1 - (1 - \frac{c_s}{c})^{c_r}$ (c being the number of possible channels, c_s and c_r the number of channels a sender/receiver is using simultaneously).

Uncoordinated Direct-Sequence Spread Spectrum (UDSSS) [63] and Randomized Differential DSSS [64] are techniques based on the same principle. They rely on the statistical chance that spread codes randomly chosen by sender and receiver(s) will happen to be the same every so often.

While the proposed methods can effectively defeat jamming and modification attacks, the inherently lower performance and a prolonged transmission time make them difficult to use in a large-scale system such as ADS-B. Furthermore, authentication and security against replay attacks is only achieved by adding a private/public key infrastructure and timestamps, respectively.

B. Public Key Cryptography

Cryptographic measures have been a tried and tested means to secure communication in wireless networks and must subsequently also be considered in the ADS-B setting. One question to examine is if the current implementation of ADS-B can be encrypted. The first possibility would be to distribute the same encryption keys to all ADS-B participants worldwide, or at least to aircraft and ground stations in a given area. Such a vast group encryption scheme, including even general aviation, would be considered extremely insecure to both inside and outside attacks. This inherent weakness is non-fixable even with

very frequent key updates (which also would again increase the complexity of the encryption deployment). In short, such a scheme would fulfill none of the required criteria listed in Section II-C. Finke *et al.* [65] examine various encryption schemes, including the possibility to do the key management for symmetric encryption out of band, for example through the controller-pilot data link communication (CPDLC) which they consider worth exploring further. The authors also give an analysis of the security and practicability of asymmetric, symmetric and format preserving encryption. In their conclusion, they support a symmetric cipher using the FFX algorithm (format-preserving, Feistel-based encryption with multiple implementation variances) which can encrypt non-standard block sizes (i.e., ADS-B's 112 bit messages) with sufficient entropy. However, the difficulties concerning key management and distribution are strongly acknowledged. Most recently, Wesson *et al.* [66] look at the broader question of how to use encryption to secure ADS-B and conclude that the problems with symmetric cryptography are too large to overcome. They argue that PKI is the only feasible cryptographic approach and propose ECDSA signatures as the smallest and thus best solution. Besides the key management problem, they further analyze the interference burden on the ADS-B channel, showing that even without the significant additional traffic that is currently found on the 1090 MHz frequency, the decrease in operational capacity would potentially be crippling.

As mentioned before, if broadcast authentication is needed, one requires an asymmetric property, a characteristic fulfilled by public key cryptography. Samuelson and Valovage [67] report on an implementation of authentication and encryption in UAT using a public key infrastructure (PKI). Their method uses a hash to create a message authentication code (MAC) that can be used to authenticate the message and can be extended to full encryption but no further details are publicly available. There are related patents filed under the names "Secure ADS-B Authentication System and Method" [68] and "Automatic Dependent Surveillance System Secure ADS-S" [69] by Sensis Corporation.¹⁶ The general idea is to use a challenge/response format with an authenticator ground station, who authenticates every participant in its reach and notifies a higher authority and/or all other participants of any failed authentication. This concept requires the station to have access to a worldwide database of secure keys that is both hard to maintain globally as well as subject to possible security breaches. If this is the case, the system can be used to not only identify ADS-B participants but also pilots and various other people/systems taking part in the flight process. Furthermore, ADS-S includes a changed modulation and a complete overhaul of the messaging system, making it incompatible with ADS-B and as such very costly and extremely doubtful to be deployed widely in the future.

Costin *et al.* [6] suggest a "lightweight" PKI solution which essentially amounts to a retroactive part publication of the key as discussed in Section IV-C: Aircraft A transmits the signature distributed over a number N of ADS-B messages, so

¹⁶Now owned by defense contractor Saab AB.

The obvious idea for a centralized key distribution would be to have aviation authorities such as the FAA act as a certificate authority (CA). But assuming the role of a CA is no easy task. Even many specialized institutions had to report numerous security breaches over the last decades. Furthermore, if this problem is sufficiently solved, there remains the question of how aircraft from airspaces mandated by different authorities can securely communicate with each other. These challenges are somewhat analog to the same approach in vehicular networks as discussed in [75] but arguable even worse due to the large internationalization of the ADS-B network.

There are certain natural disadvantages to using an encryption solution that cannot be overcome (or only with great difficulty) as mentioned in [76]:

- Despite the encryption of data frames, management and control frames are not protected.
- It immediately and unmitigatably breaks compatibility with the installed base.
- Key exchange is notoriously difficult in *ad hoc* networks, which are by definition without a centralized institution. They are often too dynamic, requiring constant adaptation. This would result in too much overhead in both the number and the size of messages.
- The open nature of ADS-B is widely seen as a feature. A cryptographic system implemented in a way comparable to ADS-S does not offer public broadcast communication.
- One-time signatures even using advanced techniques such as Merkle–Winternitz prove infeasible due to their overhead of 80 bytes and more, simply to sign 60 bits [46].

To conclude this section, it is difficult to build any kind of encryption scheme with the currently standardized 1090ES data link. Approaches have been shown to be theoretically possible with the higher bandwidth UAT, although practical proof of scalability and practicability have not been given yet. Furthermore, at this point in time it does not seem likely that UAT will play a role apart from general aviation in the FAA-mandated airspace. So, while UAT offers more technical possibilities not only for security and encryption, and even combined UAT/1090ES transmitters are neither a technical nor a regulatory problem,¹⁷ traditional cryptography in conjunction with the current installment of ADS-B seems to be a very difficult route for further research at the present.

C. Retroactive Key Publication

A variation on traditional asymmetric cryptography is the technique of having senders retroactively publish their keys which are then used by receivers to authenticate the broadcast messages. This approach that has been proposed for use in various fields [30], [77]. The key concept is simple: Any broadcasting entity produces an encrypted message authentication code (MAC) which is then sent along with every message. After a set amount of time or messages, the key to decrypt this MAC

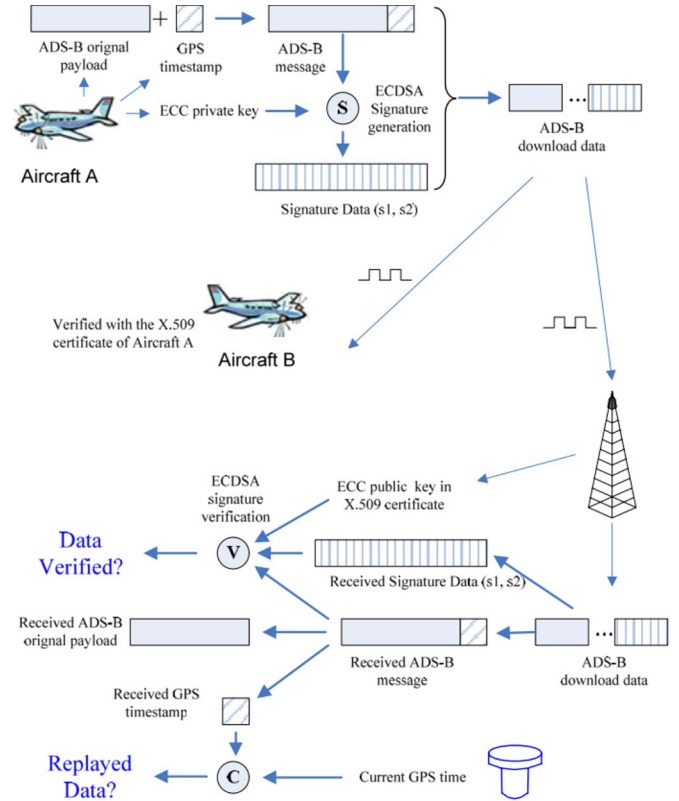


Fig. 8. Example of a typical encryption scheme adapted for ADS-B from [71]. It employs elliptic curve cryptography to generate signatures which can be verified with the responding certificate by other aircraft and ground stations. An additional GPS timestamp prevents replay attacks.

is published. All listening receivers, who have buffered the previous messages, can now decrypt the messages and ensure the continuity of the sender over time.

The TESLA (Timed Efficient Stream Loss-Tolerant Authentication) protocol [78], standardized in RFC 4082,¹⁸ can provide efficient broadcast authentication on a large scale, while it is able to cope with packet loss and real-time applications. The μ TESLA broadcast authentication protocol is the adaptation of TESLA for wireless sensor networks [79].

Both TESLA and μ TESLA use one-way key chains, as shown in Fig. 9: The broadcaster chooses a random key K_n and applies a public pseudo-random function F as often as required to acquire the keys: $K_i = F(K_{i+1})$, $0 \leq i \leq n-1$. Subsequently, every secret K_i , $i > 0$ is used for sending in the i -th interval and disclosed to the public after a number of time intervals d . As every previous key K_i with $i < j$ can be recovered by the receiver(s) by applying the one-way function F , the receiver needs to do two things to authenticate a message: [80]

- 1) Authenticate the key K_i against previously received keys to ensure they are from the same key chain.
- 2) Ensure that the message with key K_i could only have been sent before the key has been published (requiring loose time synchronization), i.e., before interval $i + d$.

¹⁷“There is nothing in the regulations and there are no technical hurdles that would prevent a manufacturer from building a combination UAT/1090ES box, one that would eliminate the different-technology blind spots while allowing high-flying aircraft the ability to get FIS-B through UAT.” <http://www.flyingmag.com/technique/proficiency/ins-and-outs-ads-b>.

¹⁸“Timed Efficient Stream Loss-Tolerant Authentication (TESLA): Multicast Source Authentication Transform Introduction,” <http://www.ietf.org/rfc/rfc4082.txt>.

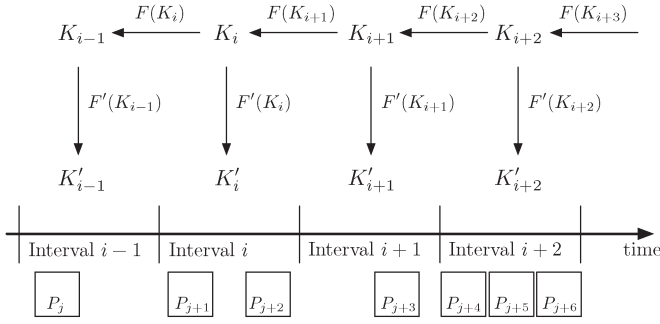


Fig. 9. The figure illustrates TESLA's utilization of one-way chains after [78]. The first one-way function F generates the chain, following that the second one-way function F' derives the MAC keys. Time is divided into separate intervals i , all having the same length. The packets P_j are each sent during one specific interval. For every such packet, the sender computes a MAC with the key that is in accordance with that interval. E.g., P_{j+2} 's MAC is calculated based on its data and key K'_i . Disclosing the keys of previous intervals can be done either by attaching the key to sent packets or in separate messages.

The fact that μ TESLA uses symmetric cryptography in connection with time as its asymmetric property makes it an interesting idea for adaptation to ADS-B since a sufficiently good time synchronization could be provided via GPS (this would require sending the GPS timestamps in a new protocol field since this is not currently the case). The advantages of μ TESLA are obvious: ADS-B keeps its open and broadcast nature and a complex PKI infrastructure is not required to ensure a sender's continuity, although it could be added if identification and source integrity are required (e.g., solely for well-connected ground stations). Nonetheless, it enables a participant to protect itself against impersonation attacks. In areas well-served by ground stations any break in continuity detected by any single one of them would set off red flags.

Another advantage of μ TESLA is that lost packets on the notoriously jammed 1090 MHz frequency (there is no medium access control in place) are not an integral problem for authentication. Furthermore, the overhead for communication as well as required modifications to the ADS-B protocol are significantly less than with traditional asymmetric cryptographic methods (see Fig. 10).¹⁹

On the other hand, μ TESLA also has some disadvantages when applying it to AANETs. There is a need to reinitialize it (if used for identification) and it can be susceptible to memory-based DoS (depending on the setup of the receiver). To counter this, Eldefrawy *et al.* [81] propose an approach that utilizes forward hashing using two different nested hashes and the Chinese Remainder Theorem, resulting in a system that does not need to be reinitialized.

Haas and Yu [82] compare TESLA and ECDSA-based authentication by simulating their performance in a real-world VANET scenario (although not including certificate distribution). Regarding channel congestion and MAC layer delay, they found that the TESLA protocol with keys attached to a subsequent broadcast performed significantly better than ECDSA or a TESLA-scheme that publishes keys in separate packets.

Hu and Laberteaux [83] discuss a combination of a full-blown PKI infrastructure with CAs distributing 512 bit sig-

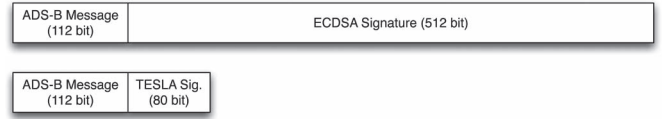


Fig. 10. TESLA signatures vs. ECDSA signatures. Tesla signatures cost significantly less overhead compared to many other cryptographic solutions.

natures to bootstrap 80 bit TESLA signatures for short-term authentication in VANETs. Such a system offers comparably lightweight integrity and possibly on-demand authentication, if needed and requested.

V. SECURE LOCATION VERIFICATION

Besides securing the communication—and thus the location data—of ADS-B, there are other approaches to ensure the integrity of air traffic management. The general idea of secure location verification is to double check the authenticity of location claims made by aircraft and other ADS-B participants. This is inherently different from the verification of the broadcast sources and messages. The baseline is to establish means to find the precise location of a sender, effectively offering some redundancy and thus the ability to double check any claims made. As an additional advantage any such approach creates more location data, which can be merged with ADS-B and radar and offer a back-up system in case of failure of these primary navigation systems or GPS.

A. Multilateration

Multilateration, or hyperbolic positioning, is a popular form of *co-operative independent surveillance* and has been successfully employed for decades in military and civil applications. If the precise distance between four or more known locations and an unidentified location can be established, it is a purely geometric task to find the unknown point. We can, for example, use the received ADS-B signals which travel at the speed of light to estimate the distance. Since we do not know the absolute time a message needed to travel from an airplane to a receiver, we have to employ the time difference of arrival (TDOA).²⁰

Thus, multilateration requires a number of antennas in different locations that receive the same signal at different times. From the TDOA, hyperboloids can be calculated on which the aircraft's position must lie. With four or more receivers, a 3D position can be estimated by finding the intersection of the hyperbolas, as shown in Fig. 11.

Performing multilateration by utilizing TDOA is currently the preferred solution for location verification on the ground. It is used in the field (e.g., by the ASDE-X system [88]) at various US airports²¹ and also being rolled out in Europe in connection with the CASCADE project.²² One major advantage of multilateration is the fact that it can utilize aircraft communication that is already in place. Thus, there are no changes required to the currently existing infrastructure in aircraft, while on the

²⁰For a full explanation of the multilateration process in aviation see, e.g., [84] or [85]. For an overview over wireless location techniques, see [86].

²¹See the manufacturer's description: <http://www.saabsensis.com/docs/128/>

²²<http://www.cascade-eu.org>

¹⁹The original μ TESLA requires a 6 byte MAC.

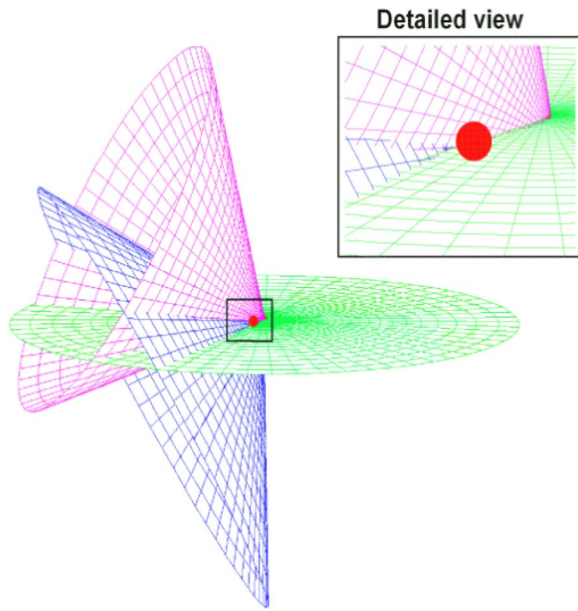


Fig. 11. Intersection of three hyperboloids from [87]. With four receivers in a 3D setting, one can specify the origin of the message as the red point where the computed hyperboloids intersect.

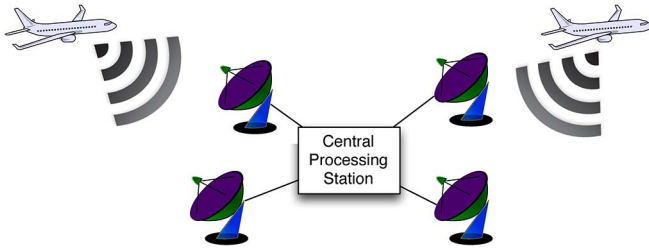


Fig. 12. Basic multilateration architecture. Four (or more) receiver stations measure the time at which they receive the same message from an aircraft. They send this data to the central processing station which can calculate the aircraft's position from the time difference of arrival between the receiver stations.

ground receiver stations and central processing stations have to be deployed (see Fig. 12).

While currently used mainly in comparably short distances (taxiway and runway on airports, up to about 60 m height), Wide Area Multilateration (WAMLAT) has also been a popular research topic. Compared to primary radar systems, WAMLAT is relatively easy and cost-effective to install and use on the ground but can also be successfully employed in an airborne MANET.²³ Using an estimated distance of a target between four or more receivers, it is possible to tell the 3D position of a sender with roughly 30 m accuracy (at 90 NM distance) compared to 20 m for ADS-B [24]. However, in comparison to ADS-B the accuracy of multilateration in practice deteriorates over long distances (see Fig. 13).

There have been various practical studies of multilateration using ADS-B signals. For example, [14] examines it as a method to provide a means to backup and validate ADS-B communication. Johnson *et al.* [89] describe their proof of concept work in a war-zone in Afghanistan. Kaune *et al.* [90] built a

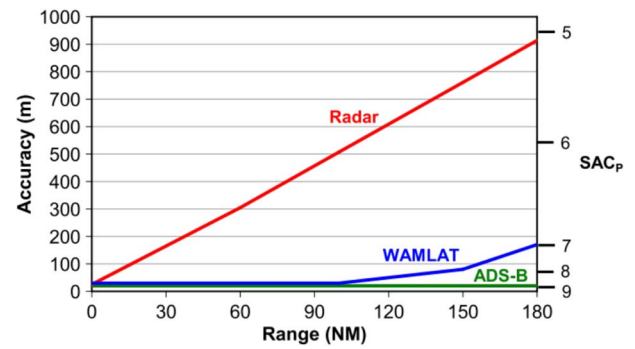


Fig. 13. Comparison of location estimation accuracies when utilizing primary radar, wide area multilateration and ADS-B [14]. SAC_p denotes the Surveillance Accuracy Category for Position as defined in [14].

proprietary low cost test bed to do multilateration with ADS-B signals. Thomas [91] presents findings from two controlled helicopter flights in the North Sea.

Daskalakis and Martone [92] give a technical assessment of the possibility of using ADS-B and WAMLAT in the Gulf of Mexico, testing a single controlled flight with good accuracy.

A recent work at MITRE Corporation [93] analyzes the attempt to build an alternative navigation system with WAMLAT in the FAA-mandated US airspace. The authors discuss the potential use of already deployed sensors for multilateration around three airports with flat terrain, also noting that the challenge is greater in more mountainous areas. Further, they provide some sensor placement discussion, determining the optimal choice and number from a given database concerning requirements such as accuracy and low dilution of precision.

Despite its successful use in the field, multilateration leaves a number of open problems in terms of secure location verification, for example the estimation of aircraft altitudes with ground-based receivers is known to be very difficult. Galati *et al.* [94] discuss the theoretical application of SSR as the basis for multilateration in airport surveillance. They analyze a case study of the Marco Polo airport in Venice and look at technical details such as dilution of precision and multilateration algorithms and conduct simulations with five sensors in a 25 km radius around the airport. They propose angle-of-arrival measurements to improve the unsatisfying height estimates provided by the wide area multilateration.

The International Civil Aviation Organization also names a few known drawbacks of multilateration: [41]

- 1) It is susceptible to multi-path propagation.
- 2) A signal has to be correctly detected at comparably many receiving stations.
- 3) A separate link between the central processing station and all receivers is required.

Essentially, some of the limitations stem from cost and logistic reasons as it can be difficult and expensive to deploy enough sensors and stations in very remote and inaccessible areas.

Furthermore, [69] mentions an attack vector for an adversary trying to fool a receiver system utilizing multilateration for location verification. An attacker would need to purchase and modify four traffic-collision avoidance system (TCAS) receivers and use a GPS/WAAS time transfer unit between the units to ensure relative timing accuracy. Furthermore, he

²³ Although vast open spaces such as found in, e.g., Australia or over oceans can prove infeasible for the use of multilateration and have been one of the very reasons driving the development and deployment of ADS-B.

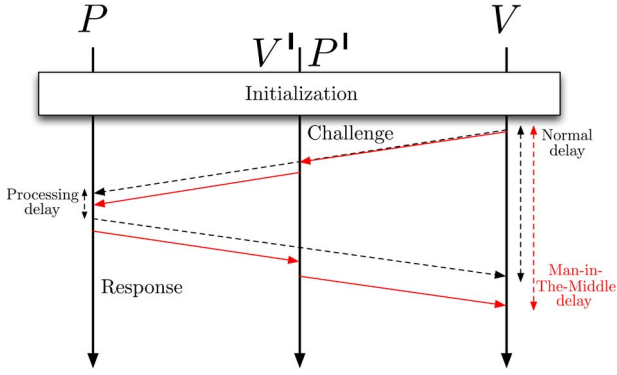


Fig. 14. Principle of distance bounding protocols. The verifier V sends a challenge to the prover P who then, after processing, sends his response (black dashed arrows). A man in the middle (V'/P') can only increase the distance by adding further processing delays, but not decrease it (red arrows).

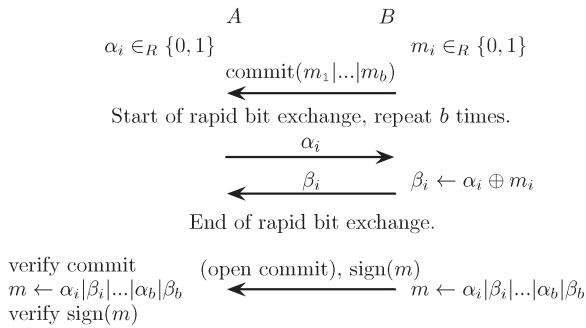


Fig. 15. Original untrusting distance bounding protocol by Chaum and Brands [96]. A is the verifier, B the prover. After the protocol exchange A can verify that B is within a given distance.

needs to engineer an algorithm similar to the one TCAS uses to determine aircraft's tracks. As this involves both a certain cost and non-zero engineering knowledge the difficulty of exploiting this threat is relatively high, certainly when compared with the simplicity of spoofing ADS-B messages.

In principle, there always remains the question of how to secure the communication needed to perform multilateration and relay the localization information to the other participants. If it is not properly secured, Sybil attacks, where a number of dishonest nodes deceive their environment, are entirely possible [95].

B. Distance Bounding

Distance bounding is another method that has been employed in wireless networks to partly localize other participants and ensure secure transactions, e.g., for RFID communication (see Fig. 14). First presented by Brands and Chaum in 1993 [96], the idea behind distance bounding is to establish a cryptographic protocol with the goal to have a prover P show to a verifier V that P is within a certain physical distance (see Fig. 15 for the concrete protocol). The universally valid fact that electromagnetic waves travel roughly at the speed of light c , but never faster, builds the foundation of all distance bounding protocols.²⁴ This enables the computation of a distance based on the time of flight between the verifier's challenge and the

corresponding response by the prover. The determined distance serves as an upper-bound, an additional piece of information that can subsequently be used as a means to verify and authenticate a node by checking the truth of its claims. When distance-bounding is performed by various trusted entities (such as ground stations) these can collaborate and find the actual location of the prover via trilateration.²⁵ There are various practical attacks on distance bounding schemes given in the literature, among them a number of relay attacks such as the so-called distance fraud, Mafia fraud and terrorist fraud [97] as well as the newer distance hijacking attack [98].

Consequently, an abundance of protocols have been suggested to deal with these various deficiencies. Song *et al.* [99] give an example of a secure distance bounding mechanism for VANETs, comprising of three steps:

- 1) Traditional distance bounding is used to find the lower bound of the distance between V and P . P can only increase the time to respond to V 's challenge and as such only appear further away than it really is.
- 2) The verifier then checks the claimed location of P for plausibility (also see Section V-F):
 - a) Transmission range-based verification: There are limits on the maximum distance of wireless transmission in practice. Test runs can help to find practical upper bounds for ADS-B users in a certain location/path. If the prover claims to be further away, it can be considered malicious.
 - b) Speed-based verification: Considering the fact that the typical speed of a given airplane in differing flight stages is known (certainly the physically possible minimum and maximum velocity), consecutive position claims have to be in a given window.
- 3) To further improve the security, after all plausibility checks have been passed, the verifier chooses a common neighbor B . That neighbor of both P and V then gives its location estimation E for P , as shown in Fig. 16. Whenever the estimate lies outside the error margin, B knows that P enlarged its distance.

Chiang *et al.* [100], [101] develop a secure multilateration scheme based on distance bounding that, under idealized assumptions, can detect false location claims with a high rate of success. By also taking into account RSS differences, they can mitigate distance enlargement attacks and generic collusion attacks on the protocol. This shows how different physical layer techniques can be combined to improve theoretical security, however, the practical challenges in using such protocols in ATC are difficult to solve.

While in the literature distance bounding has been used mostly for close-up, indoor communication, it has been modeled for use in VANETs up to a distance of 225 m between prover and verifier [99], [102]. Tippenhauer and Capkun [103] also considered the impact of moving targets on distance bounding protocols and verifiable multilateration. In their

²⁴This is in contrast to, e.g., distance estimation via received signal strength, which can be influenced and faked by a malicious node.

²⁵Triangulation is not to be confused with the previously described multilateration. The former uses the absolute measurements of three or more distance circles to determine positions, while the latter uses the *difference* in distance between the measurements.

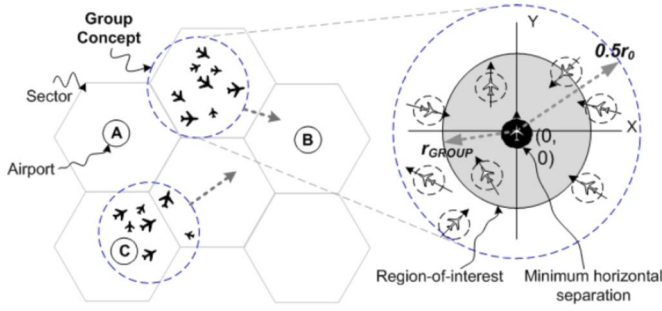


Fig. 20. Illustration of the group concept from [34]. Four or more aircraft V are in any group G . Each group then can internally use multilateration to verify each other's location claims as well as those of outsiders in range. r_0 is the wireless communication range, $0.5r_0$ considered geographically proximate and thus acceptable for group establishment, given sufficient communication quality. To lower group overhead, the region of interest for a group can be restricted to r_{group} .

From an attacker's point of view, Kalman filters can be tricked by a so-called frog boiling attack [108]: The adversary is jamming the correct signal, while continuously transmitting an ever-so-slightly modified position. If this is done slowly enough, the Kalman filter will see the injected data as a valid trajectory change. This exposes a general weakness of Kalman filtering as the approach is based on comparatively little historical data. But it is still of great use since obviously bogus maneuvers, speeds, features can be detected (see also Section V-F) and the complexity of any attack is greatly increased. Another general downside is that it opens up more DoS-possibilities due to the largely increased computational complexity at every receiver, although this is not a major problem with comparably powerful installations in ground stations and airplanes. A possible threshold time after which sufficient trust has been established between two participants based Kalman filtering is still open research [36].

D. Group Verification

Group verification is another concept proposed to mitigate security and privacy concerns over the use of ADS-B [34]. It aims at securing the airborne ADS-B IN communication by employing multilateration done by a group to verify location claims of non-group members in-flight. A given authenticated group with 4 or more aircraft having established trust can communicate with each other to utilize multilateration (based on TDOA or RSS) just as ground stations can (see Section V-A). If a forged position report is detected, the sensible reaction would be to increase the circle of avoidance around nearby airplanes since their position cannot anymore be regarded as precisely known and thus safe (see Fig. 20).

Kovell *et al.* [36] conducted a study about the applicability of the group concept in commercial aviation in the United States airspace. Examining the vast differences in traffic density over the US, they found that around 91% of aircraft at a given time could be part of a sufficiently large group of 4 aircraft or more.

Group verification has a number of downsides. First of all, it requires many additional messages to implement the verification and trust process. As ADS-B is purely unidirectional broadcast, a new protocol is needed to support the group

concept. Concerning the question of which protocol to use, the authors mention the L-Band Digital Aeronautical Communication System (L-DACS) as one possibility. L-DACS is being developed by EUROCONTROL as a future IP technology for air-to-air communication but unfortunately there is no specification in sight in the medium term.²⁶ If such a protocol can be successfully implemented, there remains the central problem of how to manage the secure authentication of members that are to be accepted into the group in the first place. It is very complicated to establish trust in new groups of MANETs and to reliably avoid malicious aircraft. Furthermore, the performance of the system in reaction to intelligent intentional jamming of some or all communication would have to be considered.

On the other hand, even without a perfectly secure solution, the group concept would raise the difficulty and engineering effort of certain airborne attacks by orders of magnitude.

E. Data Fusion and Trust Management

Data fusion is quickly becoming a cornerstone of modern intelligent transport systems (ITS). The concept can be used at various stages of data processing, Baud *et al.* [109], for example, describe the fusion of radar and ADS-B data and show that this approach can improve the quality of tracking in practice. Concerning ADS-B security, the literature proposes to check positional data obtained from within the system against data coming in from other, independent sources. Adequate data can, e.g., stem from multilateration (see Section V-A), traditional primary radar systems or even flight plan data. Liu *et al.* [110] describe a hybrid estimation algorithm to fuse multiple sensors with different surveillance techniques (PSR, SSR, multilateration) and flight plan information together, not specifically for security purposes but general fault detection.

Such verification can provide a way of knowing if some of the involved systems work outside normal parameters, be it from a malicious source or not. Subsequently, automated technical or non-technical procedures can be carried out, identifying the problem and reacting accordingly. This process comprises an analysis of the trust-worthiness of the data, if it has been vulnerable to tampering depending on the system and the precision/measurement uncertainty of the respective technologies (as given, e.g., in [14]). The trust-worthiness can then be calculated by looking at the correlations and further features deduced through machine learning processes which aim to expose anomalies in received information and thus to enable more automated detection of attacks.

An example of this is given in [111]. The authors use the cosine similarity between claimed and estimated positions to judge the trustworthiness of a participant's claims and maintain historical beacon trust information:

$$Sim_{Cos}(\vec{E}, \vec{O}) = \frac{\vec{E} \cdot \vec{O}}{|\vec{E}| \cdot |\vec{O}|} = \frac{x_E \times x_O + y_E \times y_O + v_E \times v_O}{\sqrt{x_E^2 + y_E^2 + v_E^2} \times \sqrt{x_O^2 + y_O^2 + v_O^2}} \quad (1)$$

²⁶"In addition to the air/ground capability, some of the assessed technologies could also support additional features such as air/air (point to point and/or broadcast) communications and digital voice. However the support of these capabilities needs further investigation." http://www.eurocontrol.int/communications/public/standard_page/LDACS.html.

where x_O, y_O are the coordinates, v_O the velocity as claimed in the last received message and x_E, y_E are the estimated coordinates of the claimant, based on the previously received message. As a further step they calculate the time-based weighted trustworthiness of a beacon message, taking into account the cosine similarity of the last I beacon messages and their respective estimates:

$$T_{beacon} = \frac{\sum_{i=1}^I \text{Sim}_{Cos}(\vec{E}, \vec{O})(w_i)^n}{\sum_{i=1}^I (w_i)^n}. \quad (2)$$

It is easy to make a case for data fusion, since many of the required components are already available and a two-out-of-three (2oo3) approach is a widely accepted best practice in many industries dealing with processes that are crucial to safety and security. It has even been suggested to equip the average car with primary surveillance radar, to secure vehicular *ad hoc* networks [112].

There are already a number of integrated systems being deployed (such as ASDE-X) that are inherently fusing the data of various sub-systems (ADS-B, multilateration, flight plan information, radar data) to increase both security and accuracy in airport proximity. An exemplary patent for a data fusion apparatus looking to improve ADS-B security can be found in [113]. The clear advantage of data fusion is the compatibility with legacy systems, including the fact that the ADS-B protocol does not need to be amended to provide these additional features. The downsides include increased cost for additional systems to provide the necessary redundancy.

F. Traffic Modeling and Plausibility Checks

Traffic modeling could provide a mechanism to detect deviations from normal ADS-B behavior. By utilizing historical data as well as machine learning methods it is possible to create a model of a map for each ground station, providing a means to verify location claims made by aircraft via ADS-B. Fig. 21 shows how a typical heat map based on RSS values looks from the point of view of a ground station. Such models can provide hints about non-matching location claims of a message send by an aircraft. Other potential considerations include checking for a certain number of consecutive packets of the same—or different—aircraft with the same RSS/angle-of-arrival, or otherwise suspicious absolute values which could indicate a stationary, ground-based attacker (e.g., RSS/AoA values outside typically observed thresholds).

Xiao *et al.* [114] used a similar statistical approach for the detection of Sybil nodes in VANETs. They propose an algorithm that could be performed by any node that has received enough measurements of, e.g., signal strength from nearby witnesses. While a single estimated position of a node may not be an accurate representation of the real location, a larger sample of estimated positions would have to be very similar to the node's position claims over a given period of time.

More formally, if in a period Δt_o there are n sequential positions $l_1, \dots, l_n$ claimed by an airplane with the corresponding estimated positions $l'_1, \dots, l'_n$, then the difference between the two can be treated as a random error. That means that a large enough sample of differences is distributed normally with a

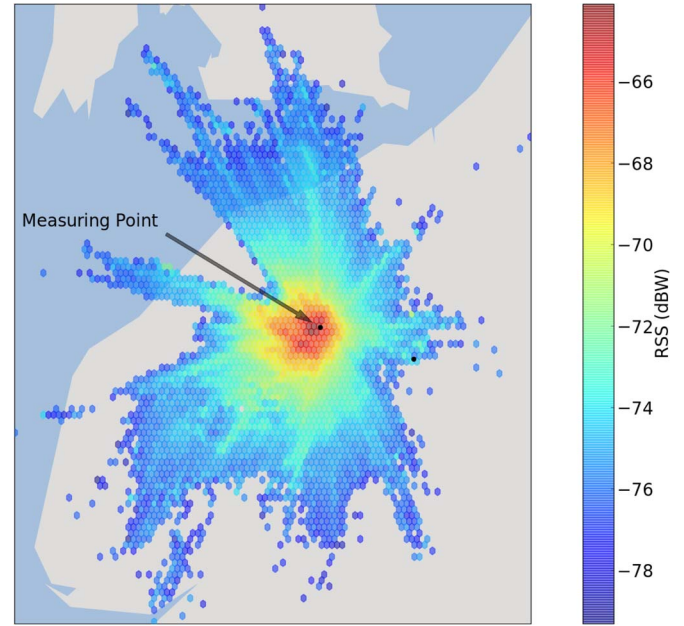


Fig. 21. Example of an RSS heat map used for traffic modeling from [9]. The colors indicate the received signal strength at a single measuring point which correlates strongly with the distance. Though it is relatively easy for an attacker to manipulate the RSS, it becomes increasingly difficult the more measuring points have to be deceived.

mean μ_d and a variance σ_0^2 . If these hypotheses are true, then it can be assumed that the claimant has given valid positional reports within the chosen level of significance (see [114] for further explanations and simulation results in VANETs). As this method is only practical for fixed receivers, the goal is to use ground stations to detect unusual behavior, which are also more likely to be able to collect enough samples to make the verification process as sound as possible.

On top of this, there are numerous comparably simple rules that can be utilized as potential red flags by an intrusion detection system without resorting to more complex measures. Neither of these rules are necessary nor sufficient by themselves to detect an ongoing attack. But depending on the scenario and the attacker's savvy, they can indicate unusual behavior that should be further investigated either by a human or handled through additional technical means. For example, it is very plausible to outright drop a number of packets where either the data or the meta data are technically or physically impossible. Doing so can significantly reduce the strain on the ADS-B system and even prevent both spoofing and DoS attacks which are not crafted carefully enough. If a large number of potential red flags are checked for an intrusion detection system, not only the risk for an attacker to cause an alert increases significantly, but also the cost and complexity of an attack rise. While this does not constitute theoretical perfect security, plausibility checks can be very useful in practice.

Mitigating factors exist across various layers, from the physical to the application. Some are available to ground stations/air traffic control only, others also to aircraft-to-aircraft (A2A) ADS-B IN communication. Such cases include but are not limited to [115]:

- Investigating **airplanes which suddenly appear** well within the maximum communication range of a receiver.

TABLE III
OVERVIEW OF CAPABILITIES OF VARIOUS SECURITY APPROACHES
AGAINST FEASIBLE ATTACKS ON ADS-B

	Injection / Modification	Eavesdropping	Jamming / Deletion
Physical Layer Authentication	+	-	-
Uncoordinated Spread Spectrum	-	+	+
(Lightweight) PKI	+	+	-
μ TESLA	+	-	-
Wide Area Multilateration	+	-	-
Distance Bounding	+	-	-
Kalman Filtering	+	-	-
Group Verification	+	-	-
Data Fusion	+	-	-
Traffic Modeling	+	-	-

- Dropping aircraft which are violating a given acceptance range threshold, producing impossible locations.
- Aircraft violating a given mobility grade threshold, producing impossible minimum or maximum velocities.
- **Maximum Density Threshold:** If too many aircraft are in a given area, ATC software will typically alarm the user.
- **Map-based Verification:** Aircraft in unusual places such as no-fly areas or outside typical airways (this might possibly be better handled at the ATC software layer).
- **Flight plan-based Verification:** Flooded/attacked ground stations are able to check ADS-B messages against the existing flight plan.
- Obvious **discontinuities in one of the 9 ADS-B state vector data fields** (also see the related Section V-C).

As explained before, such potential red flags need to be handled with utmost care and typically they can not be automated but require much additional scrutiny before any action is taken (e.g., the packet/flight is considered an attack and dropped from ATC monitors). Yet, they also enable the opportunity to follow up and activate further means to secure the airspace. For example, when using such centralized detection at the ground stations, these same stations could then destroy messages sent by the detected offenders as outlined in [116], [117].

VI. SUMMARY

Tables III–V provide a compact overview over the effectiveness of all examined solutions in combating the various proposed attacks as well as in offering advantages and disadvantages concerning the feasibility to implement each approach in the real world. As it has been laid out, there is no single optimal or even good solution when considering means that have no or little impact on the currently employed ADS-B software and hardware. Table III shows the attacks the discussed approaches can counteract. We see that most security schemes focus on attacks of the message injection/modification class. This has two main reasons that have been mentioned throughout this survey: First of all, the open nature of ADS-B has been considered a desirable feature in most scenarios. So unless there is a major paradigm shift in the way air traffic communication and control is handled currently, there is no interest in protecting against passive listeners, despite this being

TABLE IV
OVERVIEW OF SECURITY FEATURES OF VARIOUS
APPROACHES FOR USE WITH ADS-B

	Data Integrity	Source Integrity	Location Integrity	DoS
Physical Layer Authentication	No	Yes	Possibly	Partly
Uncoordinated Spread Spectrum	No	No	No	Yes
(Lightweight) PKI	Yes	Yes	Yes	Partly
μ TESLA	No	Yes	No	No
Wide Area Multilateration	No	No	Yes	No
Distance Bounding	No	No	Partly	No
Kalman Filtering	No	Partly	Partly	No
Group Verification	No	Possibly	Yes	No
Data Fusion	No	Partly	Yes	Backup
Traffic Modeling	No	No	Yes	No

the first stepping stone for more sophisticated and problematic attacks. Second, passive attacks such as eavesdropping are simply much more difficult to protect against without having a full cryptographic solution. Similarly, attacks on the physical layer, such as continuously jamming the well-known frequency or the more surgical message deletion are hard to defend against, with measures on the same layer (e.g., uncoordinated spread spectrum) providing some of the only approaches to this general wireless security problem. All discussed approaches do however address message insertion and tampering, either by protecting outright against it through verification (cryptographic methods) or by detecting anomalies in the data (e.g., Kalman filtering, multilateration).

Table IV takes a look at the security features the discussed schemes can provide. As discussed before, only a full cryptographic public key infrastructure can guarantee the integrity of received data. All other approaches either aim to secure the integrity of the source (e.g., μ TESLA, many of the discussed physical layer schemes) or seek to verify the provided location data independently. Additional protection against flood denial of service attacks against ATC systems can be directly provided by spread spectrum approaches and cryptography, while other methods rely on higher layers to sort out false aircraft claims.

Table V provides an overview over the feasibility of the different approaches in practical settings, especially considering the current state of air traffic control in the aviation industry. As is to be expected, the difficulty and cost columns are mostly correlated. The difficulty to overcome technical challenges is particularly high for distance bounding, which is yet in its beginnings and a full-blown public key infrastructure. In contrast, we see wide-area multilateration, Kalman filters and data fusion techniques already in use in the field. This naturally translates to the cost factor which plays an important role in industry decisions. One can choose between a completely new protocol which addresses the security question better than the current installment of ADS-B does, slight modifications such as new message types, or a transparent, parallel system which requires new software and/or new hardware in different scales. This touches also on the question of scalability. For example, the fusion of various ATC systems and their data (PSR, SSR, ADS-C, WAMLAT, FANS) is an obvious and necessary idea. Yet, it is common knowledge in the aviation community that

TABLE V
OVERVIEW OF FEASIBILITY ATTRIBUTES OF VARIOUS APPROACHES FOR USE WITH ADS-B

	Difficulty	Cost	Scalability	Compatibility	References
Physical Layer Authentication	Variable	Variable	Variable	Requires additional hard-/software. No modifications to the ADS-B protocol.	[48]–[61]
Uncoordinated Spread Spectrum	Medium	Medium	Medium	Requires new hardware and a new physical layer.	[62]–[64]
(Lightweight) PKI	High	High	Medium	Distribution infrastructure and changes in protocol and message handling needed.	[6], [65]–[71], [73]–[76]
μTESLA	Medium	Medium	High	New message type required for key publishing, MAC added.	[30], [77]–[83]
Wide Area Multilateration	Low	Medium	Medium	No change to ADS-B required. Separate hardware system.	[14], [24], [84], [85], [89]–[94]
Distance Bounding	High	Medium	Low	New messages and protocol needed.	[96], [99]–[103]
Kalman Filtering	Low	Low	High	No additional messages needed. Separate software system.	[36], [104]–[107]
Group Verification	High	Medium	Low	New messages and protocol needed.	[34], [36]
Data Fusion	Low	Medium - High	Medium	No change in ADS-B required. Separate system.	[109]–[113]
Traffic Modeling	Medium	Low	High	Additional, separate entities for ground stations needed.	[9], [114], [115]

a “major part of the business case for Automatic Dependent Surveillance—Broadcast (ADS-B) is attributed to the savings generated by decommissioning or reducing reliance on conventional radar systems” [14]. Thus, it seems inefficient and unlikely to have legacy and/or new backup systems around on a broad scale, simply to fix the inadequateness of ADS-B related to security.

A. Future Research Directions

Considering the fact that the invention, certification and large-scale deployment of air-traffic systems takes decades, as currently seen in the example of ADS-B, it seems equally non-sensible to present a completely overhauled ATC-system at this point in time. Yet, future aviation protocol development is important as there will eventually be a successor to ADS-B and the responsible community must learn from the ADS-B case study. Authentication should be considered right from the beginning when planning new protocols, this includes choosing the right cryptographic primitives, an appropriate communication pattern considering A2A and ground stations, and, most importantly, a solution to the key management problem. Furthermore, the challenges and realities of communication in the avionic environment need to be taken into account, for example the extremely lossy environment (as recently examined and illustrated with OpenSky, an open research sensor network in [22], [118]) that might render many traditional approaches infeasible.

For the urgent problem at hand, this means that incremental changes with backwards compatibility kept in mind as a main factor are more useful than completely new proposals. Improvement on transparent secure location verification approaches such as multilateration can help bridge the security gap in the very near future, although this means losing some of the advantages for which ADS-B was originally developed. In the same vain, fingerprinting methods on various layers can help build an effective intrusion detection system against all but the most sophisticated attackers without affecting the protocol as

it is currently deployed. Improvements on current data fusion algorithms can also further both safety and security of currently deployed ATC by reducing error margins and uncertainties for controllers.

Of course, cost and complexity of deployment cannot be the only factors taken into consideration—not having security is famously even much more expensive.²⁷ As mentioned throughout this survey, it can be a useful dichotomy to distinguish between attack detection, attack prevention and dealing with suspected attacks. We focused mainly on attack detection and prevention, leaving attack reaction for future research.

VII. CONCLUSION

This survey sought to review the available research on the topic of securing the ADS-B protocol in particular and air traffic control communication in general. We provided an in-depth overview of the existing work, both specific to ADS-B as well as ideas brought in from related fields such as VANETs. After reviewing the literature, it seems that the solutions currently under consideration (and in use in practice such as multilateration) can only be a fill-in, providing a quick improvement to the security of the current system. For all-encompassing security (and possibly privacy), new message types and/or completely new protocols would need to be defined. Taking this into account for the creation of a long-term security solution in dependent air traffic surveillance, it makes sense to consider the impact of both secure broadcast authentication approaches as well as of secure location verification. To avoid new hard challenges in the foreseeable future, this should include a thorough analysis of the predicted traffic density on today’s wireless navigation channels as well as the possible impact of the communication and message overhead of a new protocol.

²⁷“If you think safety is expensive, try an accident,” an insight by easyJet owner Stelios Haji-Ioannou that came after facing manslaughter charges for the deaths of employees in a shipping disaster at a former company.

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